

Critical Capacity Completion

Equivalent Forms of the Riemann Hypothesis

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Abstract

A frozen source-derived arithmetic construction is connected to the classical Riemann problem through exact finite and infinite statements. Scalarization of the uncollapsed response algebra fails only through repeated-prime and over-capacity interactions; differentiation gives an exact boundary-current reconstruction. A weighted Mertens capacity energy A_N is introduced and proved to satisfy $A_N/N \rightarrow \infty$ and $N/A_N \rightarrow 0$. The power-growth exponent of the excess provides a critical coordinate. After classical identification, zero transverse escape, local-uniform Mobius control, completed-incidence positivity, and subpower capacity excess are proved equivalent to the Riemann Hypothesis in the displayed normalizations. The equivalence theorem is stated boldly and separately from the still-unproved universal estimate required to settle the conjecture. The static five-class Möbius reporter is proved to be a false closure under prime multiplication; the all-scale predictive quotient is the prime-support address plus one absorbing repeated state.

Keywords: Riemann Hypothesis; Mobius function; Mertens function; capacity energy; operator inequality; critical exponent

Scope and status: The finite reconstruction, weighted-energy divergence, and equivalence theorems are asserted with their displayed proofs and scopes. The universal RH-equivalent lower bound remains open; no proof of RH is claimed.

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1 Source freeze and classical identification

The upstream construction is frozen before classical comparison: binary capacity, transparent arities, independent prime coordinates, squarefree incidence orientation, quota topology, nonbinary contrast fibres, null half-transport, and the orientation operator. Only then are the generated scalar character and sign identified with the classical Dirichlet and Mobius objects in their absolute domain.

This paper studies three exact objects: the finite scalarization defect, a weighted Mertens capacity energy, and equivalent formulations of the Riemann Hypothesis. Equivalence is not itself a proof of the conjecture.

2 Finite scalarization defect

The finite uncollapsed response group is invertible before scalarization. Scalar multiplication fails for two exact reasons:

1. repeated-prime collisions;
2. products crossing the shared capacity boundary.

Let $\Phi_N(a; s)$ be the scalarized response and $\lambda_N(s)$ its neutral generator. Differentiation gives a boundary current $R_N(a; s)$.

Theorem 2.1 (Boundary-response reconstruction). *For every finite capacity N ,*

$$\Phi_N(a; s) = e^{a\lambda_N(s)} \left[1 + \int_0^a e^{-t\lambda_N(s)} R_N(t; s) dt \right]. \quad (1)$$

Every term in R_N is generated by a repeated-coordinate or over-capacity interaction.

Finite zeros therefore satisfy an exact coupled charge-current constraint. The theorem identifies the missing scalar information; it does not estimate the infinite signed current.

3 Weighted Mertens capacity energy

Let

$$M(x) = \sum_{n \leq x} \mu(n), \quad (2)$$

and define

$$A_N = \sum_{q \leq N} M\left(\left\lfloor \frac{N}{q} \right\rfloor\right)^2, \quad \lambda_N = A_N^{-1}. \quad (3)$$

Grouping equal values of $\lfloor N/q \rfloor$ gives

$$\frac{A_N}{N} = \sum_{k=1}^N M(k)^2 \frac{\lfloor N/k \rfloor - \lfloor N/(k+1) \rfloor}{N}. \quad (4)$$

For fixed k , the coefficient tends to $1/[k(k+1)]$.

Theorem 3.1 (Divergence of the weighted Mertens energy).

$$\sum_{n \geq 1} \frac{M(n)^2}{n(n+1)} = \infty. \quad (5)$$

Consequently,

$$A_N/N \rightarrow \infty, \quad N\lambda_N \rightarrow 0. \quad (6)$$

Proof sketch. If the weighted series converged, then $g(u) = e^{-u/2}M(e^u)$ would lie in $L^2(0, \infty)$. Its Laplace transform would have uniformly bounded vertical L^2 norms in the right half-plane. Summation by parts identifies the transform with $[(z + 1/2)\zeta(z + 1/2)]^{-1}$ in the initial domain. A critical-line zero gives boundary growth incompatible with the uniform bound. The lower limit for A_N/N then dominates arbitrarily long partial sums of the divergent weighted energy. $\square$

Table 1: Independent finite audit of the capacity energy.

N	$M(N)$	A_N	A_N/N
10	-1	11	1.100000
100	1	143	1.430000
10^3	2	1479	1.479000
10^4	-23	15829	1.582900
10^5	-48	157055	1.570550
10^6	212	1645995	1.645995

Finite nonmonotonicity is compatible with the asymptotic theorem.

4 Capacity recurrence exponent

Define the critical excess and its power-growth exponent by

$$C_N = A_N/N, \quad (7)$$

$$\chi_{\text{cap}} = \limsup_{N \rightarrow \infty} \frac{\log C_N}{\log N}. \quad (8)$$

The unconditional result is $C_N \rightarrow \infty$. The critical distinction is subpower growth versus positive-power growth.

5 Equivalent completion statements

After the classical identification, the following formulations are equivalent to the Riemann Hypothesis within the frozen normalizations used here:

1. local-uniform boundedness of the Mobius polynomials $F_N(s) = \sum_{n \leq N} \mu(n)n^{-s}$ on every compact subset of $\Re s > 1/2$;
2. a completed-incidence positivity inequality on the pole-neutral subspace;
3. absence of positive transverse escape in the normalized causal-tail representation;
4. the critical capacity condition $\chi_{\text{cap}} = 0$.

Theorem 5.1 (Equivalence theorem). *The displayed capacity-completion condition is equivalent to the classical Riemann Hypothesis.*

Claim status. Exact equivalence. The universal capacity estimate required to conclude the condition is not proved here.

6 Completed-incidence formulation

Let D_L be the completed incidence operator and $\Delta(L)$ the common budget in the frozen Weil normalization. On the pole-neutral space $\mathcal{H}_L^0$, the target inequality is

$$A_L|_{\mathcal{H}_L^0} \geq \Delta(L)I \quad \text{for every } L. \quad (9)$$

Equivalently, when $\Delta(L) > 0$, there is a contraction T_L such that

$$\sqrt{\Delta(L)} g = T_L D_L g, \quad \|T_L\| \leq 1. \quad (10)$$

The operator and scale-consistency identities are constructed; the universal lower bound is the open RH-equivalent statement.

7 Causal-tail formulation

Let Θ_N be the normalized adverse tail over degree at most N beyond a sublinear observer boundary. Define

$$\chi_{\text{RH}} = \limsup_{N \rightarrow \infty} \frac{1}{N} \log(1 + \Theta_N). \quad (11)$$

Positive χ_{RH} represents exponentially coherent adverse witnesses across increasing degree. The critical regime is the absence of positive exponential escape. This is not identical to the Schatten boundary $\Re s = 1/2$ without the explicit intertwiner supplied by the causal-tail construction.

8 Failed bridges retained

The following routes are not used as proof:

- finite zero audits;
- symmetry of the functional equation alone;
- random-sign or square-root-cancellation assumptions;
- a fitted spectrum or random-matrix analogy;
- the unbounded uniform odd trace on the Hilbert-Schmidt carrier;
- finite positivity without a uniform bound.

They remain valuable countermodels because they identify exactly where a plausible shortcut loses the required scalar orientation.

9 Static five-class reporter and all-prime predictive closure

The digit carrier admits the exact terminal-resolved classes

$$\{0\}, \quad \{1\}, \quad \{2, 3, 5, 7\}, \quad \{6\}, \quad \{4, 8, 9\}, \quad (12)$$

corresponding to zero, unit, Möbius value -1 , value $+1$, and value 0 . This is the coarsest static reporter on the ten digits that preserves terminal role and Möbius value. It is not an autonomous all-scale arithmetic state.

Theorem 9.1 (Five-class false closure under prime multiplication). *When multiplication by arbitrary primes is admitted, the five-class reporter is not a continuation congruence. The coarsest predictive quotient for all future Möbius reports consists of one state for every finite squarefree prime-support set and one absorbing repeated-prime state.*

Proof. The states 2 and 3 share the prime class. Multiplication by 2 sends them to 4 and 6, whose Möbius values are 0 and +1. Hence the static class is not future-complete. Represent a squarefree integer by its finite prime support and every nonsquarefree integer by $\dagger$. Multiplication by p adjoins p if absent and enters $\dagger$ if present; $\dagger$ remains absorbing. If two supports differ, multiplying by a prime in their symmetric difference makes one repeated and the other squarefree. Thus distinct supports are future-distinguishable and the quotient is coarsest. $\square$

The source has complete vector and mean-square realizations above $\Re s = 1/2$, while fixed-phase scalar evaluation remains unbounded. The exact RH gate is therefore not another finite classification. It is local-uniform boundedness of the special Möbius diagonal on every compact subset of $\Re s > 1/2$.

10 Conclusion

The arithmetic programme supplies a source-ordered derivation, an exact finite response reconstruction, an unconditional coercivity theorem, and several exact equivalent forms of RH. It does not yet supply the universal capacity-completion estimate. The live mathematical object is therefore precise: prove subpower critical excess or the equivalent completed-incidence bound without importing the desired cancellation.

11 Weighted capacity energy in detail

Let $M(x) = \sum_{n \leq x} \mu(n)$ and define

$$A_N = \sum_{q \leq N} M\left(\left\lfloor \frac{N}{q} \right\rfloor\right)^2, \quad \lambda_N = A_N^{-1}. \quad (13)$$

Grouping equal quotient values yields

$$\frac{A_N}{N} = \sum_{k=1}^N M(k)^2 \frac{\lfloor N/k \rfloor - \lfloor N/(k+1) \rfloor}{N}. \quad (14)$$

For fixed k , the coefficient tends to $1/[k(k+1)]$.

Suppose, for contradiction, that

$$\sum_{n \geq 1} \frac{M(n)^2}{n(n+1)} < \infty. \quad (15)$$

Then $g(u) = e^{-u/2} M(e^u)$ is square integrable. Its Laplace transform has uniformly bounded vertical L^2 norm in the open right half-plane. In the initial domain, summation by parts identifies it with the reciprocal zeta factor divided by the Mellin variable. A zero on the critical line produces boundary blow-up incompatible with that uniform bound. Hence the weighted Mertens energy diverges. Fatou's lemma applied to the grouped expression gives

$$\frac{A_N}{N} \longrightarrow \infty, \quad N\lambda_N \longrightarrow 0. \quad (16)$$

The finite audit is

N	$M(N)$	A_N	A_N/N
10	-1	11	1.100000
100	1	143	1.430000
10^3	2	1479	1.479000
10^4	-23	15829	1.582900
10^5	-48	157055	1.570550
10^6	212	1645995	1.645995

Finite nonmonotonicity is compatible with the asymptotic theorem.

12 Three equivalent critical coordinates

The classical target can be written through several exact equivalences after the arithmetic identification is frozen.

1. Local-uniform boundedness of the Möbius Dirichlet polynomials on compact subsets of $\Re s > 1/2$.
2. A completed-incidence positivity or coercivity inequality on every finite scale.
3. Zero power-growth exponent

$$\chi_{\text{cap}} = \limsup_{N \rightarrow \infty} \frac{\log(A_N/N)}{\log N} = 0. \quad (17)$$

The unconditional divergence $A_N/N \rightarrow \infty$ and the critical condition $\chi_{\text{cap}} = 0$ are consistent: the latter permits divergent growth slower than every positive power.

13 Finite scalarization reconstruction

Before scalarization, the squarefree response group is everywhere invertible. Scalar multiplication fails only when prime coordinates repeat or when a product crosses the shared capacity boundary. Differentiation at the neutral response gives a boundary current $R_N(a; s)$ and exact reconstruction

$$\Phi_N(a; s) = e^{a\lambda_N(s)} \left[1 + \int_0^a e^{-t\lambda_N(s)} R_N(t; s) dt \right]. \quad (18)$$

Every finite scalar zero therefore obeys a coupled charge–current equation whose terms are generated by those two failures. Passing from that exact finite equation to an infinite signed completion is the live analytic problem.

14 Countermodels and proof boundary

The source programme tested and retained countermodels to several tempting shortcuts: finite positivity without uniform control; quadratic completion without an odd scalar; exact reflection without a magnitude estimate; sparse bad modes without control of coherent near-resonance; and numerical zero alignment without a source-generated universal inequality. The paper proves equivalence and several auxiliary theorems. It does not claim the open universal bound.

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